

Visvesvaraya Technological University Belagavi



Mini Project Synopsis presentation on

“Web Server Implementation on ESP32 Board”

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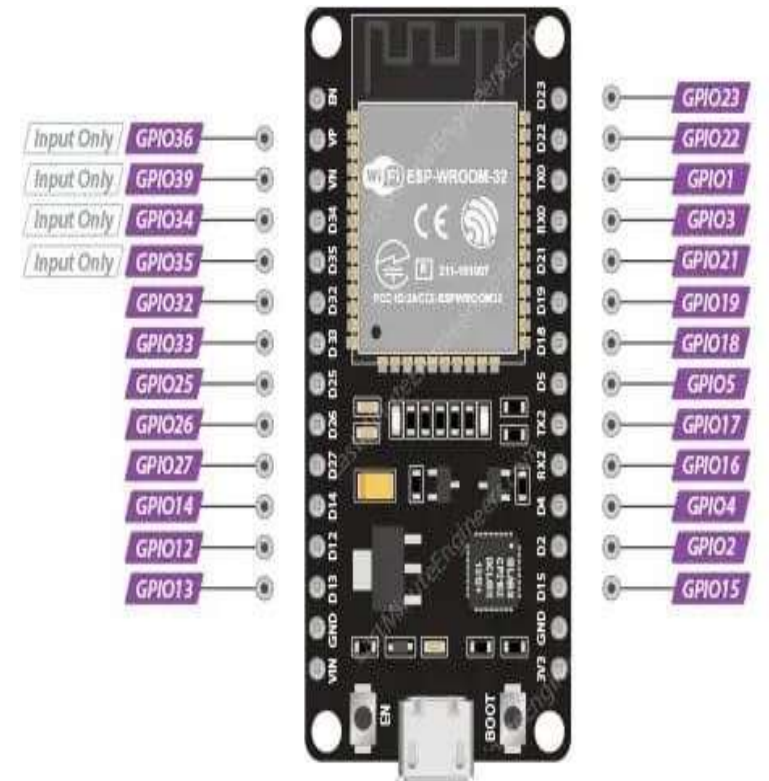
Presentation Flow

- Introduction
- Literature Survey
- Aim of the Project
- Objectives & Methodology
- Pin diagram
- Implementation Details
- Hardware/Software requirements
- Results & Discussions
- Action Plan
- References

Introduction

The ESP32, a low-cost, low-power microcontroller with built-in Wi-Fi and Bluetooth capabilities, has become a popular choice for IoT (Internet of Things) applications. One of the most common uses of the ESP32 is to implement a web server, allowing users to remotely control and monitor devices via a web interface. This can be applied to smart home systems, industrial automation, and various DIY projects.

A web server on an ESP32 allows communication between the ESP32 and a web client, such as a browser on a computer or mobile device. The web server can host a web page and respond to HTTP requests. This enables real-time control and feedback, such as switching lights on and off, reading sensor data, or sending commands to the ESP32 via the network.



Literature Survey

- **Maier, A. Sharp, Y . Vagapov** , “Comparative analysis and practical implementation of the ESP32 microcontroller module for the internet of things” 2017 Internet Technologies and Applications, ITA 2017 - Proceedings of the 7th International Conference IEEE, pp 143-148, November 2017, DOI:10.1109/ITECHA.2017.8101926.

Abstract: This paper discusses the Espressif Systems latest product ESP32 designed for Internet of Things and embedded system related projects. The ESP32 is a low-cost, low-power system on a chip series of microcontrollers with Wi-Fi and Bluetooth capabilities and a highly integrated structure powered by a dual-core Tensilica Xtensa LX6 microprocessor. This paper provides a comparative analysis of the ESP32 with some other market competitors and introduces the microcontroller specification, features and programming details. A portable, wireless oscilloscope based on the ESP-WROOM-32 and a mobile application is described in detail as an example of successful practical implementation of the device.



- **I. Allafi, T. Iqbal**, “Design and implementation of a Low cost web server using ESP32 for real-time photovoltaic system monitoring” 2017 IEEE Electrical Power and Energy Conference, EPEC 2017, pp 1-5, February 2018, DOI: 10.1109/EPEC.2017.8286184.

Abstract: This paper presents a method for applying a web server based on ESP32 to a small photovoltaic (PV) power system to monitor and/or collect the PV and the battery current, voltage data. The designed system uses low-cost sensors, a microcontroller ESP32, Wi-Fi, and an SD-card reader. The ESP32 collects data from sensors. All of the data are saved in a text file on an SD-card, which is connected with ESP32 SPI pins. The text file is saved on the SD card for a week or longer, after which the system deletes all the data and starts saving new data. The web page file is saved on an SD card as well, so the ESP32 is programmed to access the web page by using the Wi-Fi via a laptop, cellphone, or tablet. Moreover, the web page has a link that allows users to download the data text file simply by clicking on the link. This can also be done remotely. The results of the experiments demonstrate that the web server works in real-time and can be effectively used for monitoring small solar energy systems

- **B.S. Sarjerao, A. Prakasarao**, “A Low Cost Smart Pollution Measurement System Using REST API and ESP32” 3rd International Conference for Convergence in Technology, I2CT 2018, November 2018, DOI: 10.1109/I2CT.2018.8529500.

Abstract: The ongoing trends in technology enable the communication between devices through wired or wireless medium. The Internet of Things (IoT) and Machine to Machine (M2M) communication are such trends in technology which make the networked devices to communicate and performs the action without human interventions. Moreover, one problem is debilitating to air quality and its measurement in Indian cities. The major contributors to air pollution are the smog, sulfur dioxide, nitrogen oxides, CO, PM10, PM2.5. These gases are found as the most important in terms of human health Problems. It can cause lung cancer, asthma, breathing problems. In this proposed system, MQ-9 is used for measuring CO, LPG, CH4 value, DHT22 is used for measuring temperature in degree celsius and humidity in percentage, dust sensor is used for measuring dust particles MQ-135 is used for measuring benzene, ammonia, sulfur dioxide in the Environment. This measured data is put on to the server by using REST protocol (REpresentational State Transfer protocol) and ESP32. All sensors readings are displayed on the webpage and mobile application

- **A.H. Abdullah, S. Sudin, M.I.M. Ajit, F.S.A. Saad, K. Kamaruddin, F. Ghazali, Z.A. Ahmad, M.A.A. Bakar,** “Development of ESP32-based Wi-Fi Electronic Nose System for Monitoring LPG Leakage at Gas Cylinder Refurbish Plant” 2018 International Conference on Computational Approach in Smart Systems Design and Applications, August 2018, DOI: 10.1109/ICASSDA.2018.8477594

Abstract: The safety of industrial and residential environments depends heavily on the early detection of hazardous gas leaks, particularly liquefied petroleum gas (LPG), which is highly flammable. This paper presents the development of an electronic nose (e-nose) system based on the ESP32 microcontroller platform for monitoring LPG leakage in a gas cylinder refurbishment plant. The system integrates a Wi-Fi-enabled sensor array that detects the specific chemical signature of LPG, enabling real-time monitoring and prompt detection of leaks. The system is designed to be cost-effective, energy-efficient, and capable of providing early warnings to prevent potential safety hazards. A comprehensive evaluation of the system's performance, including sensitivity, response time, and reliability, is conducted in an actual industrial setting. The results demonstrate that the ESP32-based e-nose system is an effective tool for monitoring LPG leakage, offering a promising solution for enhancing safety protocols in gas handling facilities.

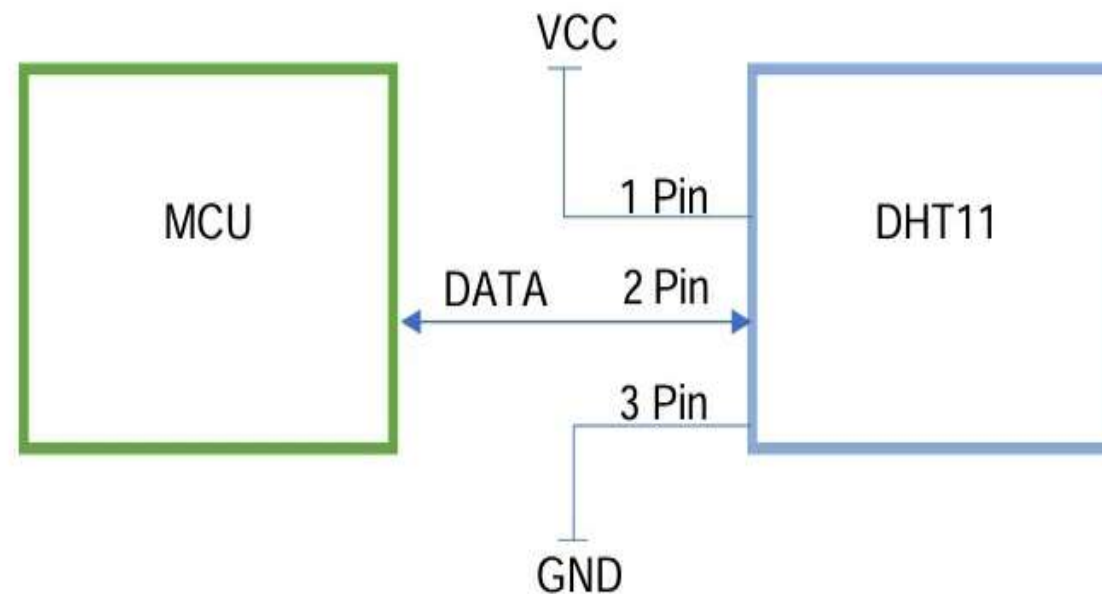
Aim of the Project

- Aim : Designing a Standalone Asynchronous ESP32 Web-Server for Temperature and Humidity Monitoring

Objectives & Methodology

Objective	Methodology
HUMIDITY AND TEMPERATURE SENSING	• MODULE SELECTION • PROGRAM SELECTION • TESTING AND REFINEMENT
WEB PAGE CREATION AND HUMIDITY AND TEMPERATURE GAUGE SCALE CREATION	• ARDUINO IOT CLOUD • DESINGINING TEMPERATURE AND HUMIDITY GAUGE • TESTING
PREPARING AND HOUSE MODLE	• BULIDING A MODLE HOUSE

Block Diagram

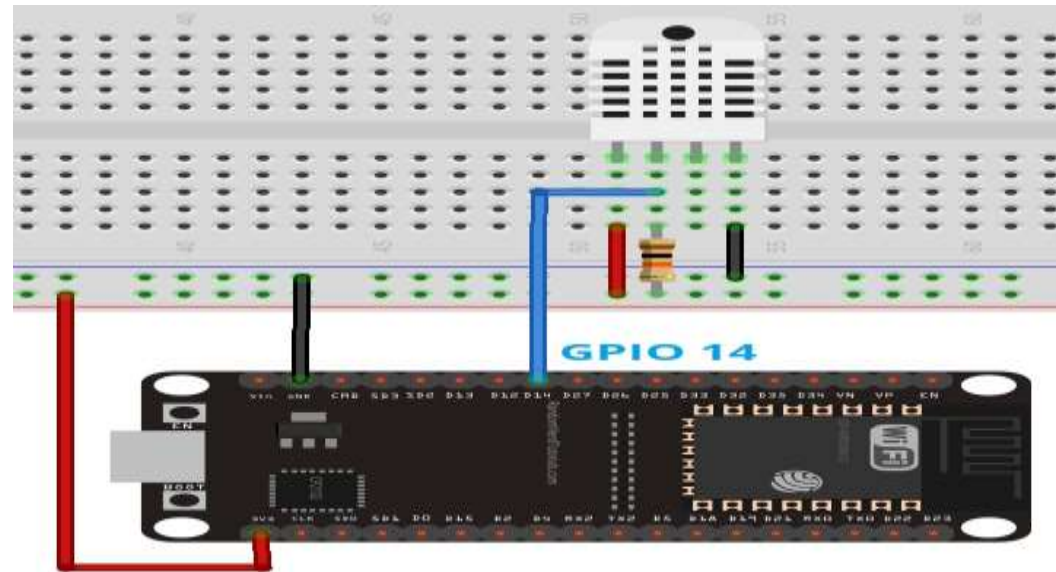


Block Diagram Explanation

- ESP32 microcontroller is connected to DHT22 sensor to transfer data from pin 2 of DHT22 sensor.
- PIN 1 is connected to VCC.
- PIN 3 is connected to GND.

Hardware Components

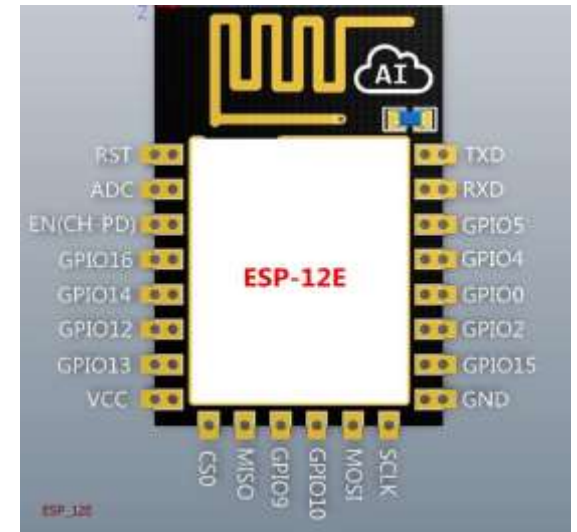
- Circuit Diagram



- The circuit diagram of an ESP8266 temperature and humidity monitoring system features the ESP8266 microcontroller at its core, interfacing with a DHT11 or DHT22 sensor to measure environmental conditions. The microcontroller processes the sensor data and transmits it via Wi-Fi to a cloud server or local network for remote access. Power supply components ensure proper voltage, while optional displays and actuators can enhance user interaction and automation, making the system versatile for environmental monitoring applications.

Hardware Components

- **Individual module explanation/introduction**
- **ESP8266 12-E** – read **Best ESP8266 Wi-Fi Development Boards**-The ESP8266 is a popular low-cost Wi-Fi microchip with full TCP/IP stack and microcontroller capability. It is widely used in Internet of Things (IoT) projects, providing easy Wi-Fi connectivity for a variety of devices. The ESP8266-12E is one of the most popular versions of this chip, often found on development boards that simplify its use for DIY projects and prototyping.
- **DHT22 sensor**-The DHT22 is a popular digital temperature and humidity sensor, commonly used in various IoT (Internet of Things) and weather-monitoring applications. It's known for its reliability and ease of use with microcontrollers such as the ESP8266 or Arduino.



- **Bread Board**-A breadboard is a crucial tool in electronics and prototyping, allowing you to build and test circuits without the need for soldering. It's often used by hobbyists, engineers, and students to quickly assemble temporary circuits and experiments. The name "breadboard" originates from the early days of electronics when inventors used wooden breadboards to build circuits.

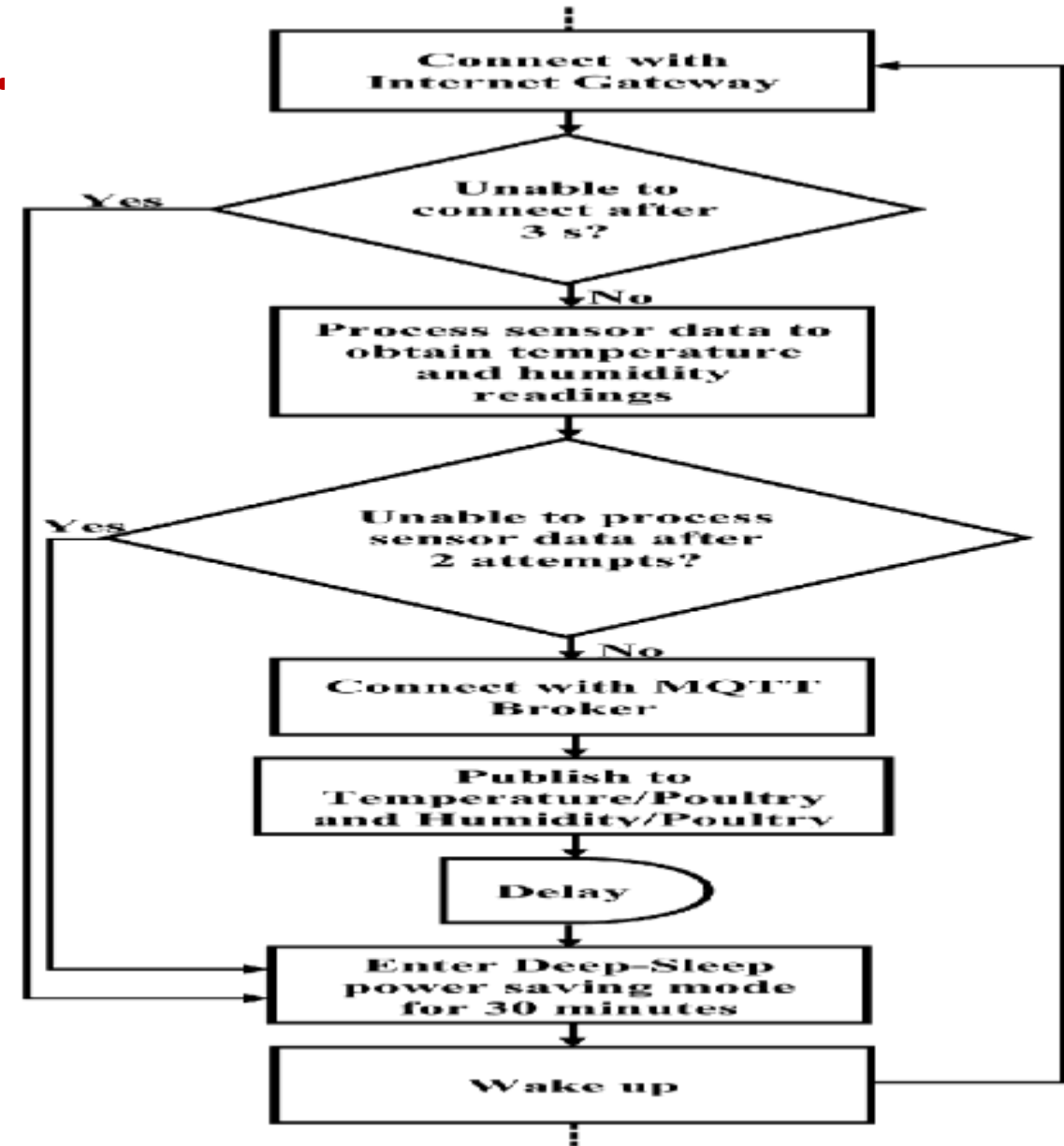


- **Jumper wires**-Jumper wires are one of the most essential tools in electronics prototyping and circuit building. These flexible wires are used to connect different components on a breadboard or between a breadboard and a microcontroller (like an Arduino, ESP8266, or Raspberry Pi) without the need for soldering.



Software Components.

- Flow Chart



- **Program**

```
#include <WiFi.h>
#include <ESPAsyncWebServer.h>

const char* ssid = "ELDRADO"; // CHANGE IT
const char* password = "amazon123"; // CHANGE IT

AsyncWebServer server(80);

void setup() {
  Serial.begin(9600);
```

```
// Connect to Wi-Fi
WiFi.begin(ssid, password);
while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.println("Connecting to WiFi...");
}
Serial.println("Connected to WiFi");

// Print the ESP32's IP address
Serial.print("ESP32 Web Server's IP address: ");
Serial.println(WiFi.localIP());
```

```
// Define a route to serve the HTML page
```

```
server.on("/", HTTP_GET, [](AsyncWebServerRequest* request) {  
    Serial.println("ESP32 Web Server: New request received:"); // for debugging  
    Serial.println("GET /"); // for debugging  
    request->send(200, "text/html", "<html><body><h1>Hello, ESP32!</h1></body></html>");  
});
```

```
// Start the server
```

```
server.begin();  
}
```

- Web Server



The screenshot shows a web browser interface for the "ESP8266 DHT Server". The address bar displays the IP address "192.168.1.95". The page content includes a heading "ESP8266 DHT Server", a temperature reading of "21.00 °C", and a humidity reading of "91.10 %". Annotations with red and green arrows point to specific elements: the IP address, the heading, the temperature and humidity paragraphs, and the placeholders "%TEMPERATURE%" and "%HUMIDITY%" within the data fields.

Annotation	Target Element
ESP8266 IP address	192.168.1.95
Heading	ESP8266 DHT Server
Paragraph	Temperature 21.00 °C
%TEMPERATURE% placeholder	21.00
Paragraph	Humidity 91.10 %
%HUMIDITY% placeholder	91.10

- To build the web server we'll use the **ESPAsyncWebServer** library that provides an easy way to build an asynchronous web server. Building an asynchronous web server has several advantages. We recommend taking a quick look at the library documentation on its **GitHub** page

Results Obtained

- We have created web cloud server through ARDUINO IOT cloud.

Action Plan:

Sl. No.	Tasks to be performed	Deadline (Expected date/week of completion)
1.	Literature Survey	
2.	Synopsis Submission	
3.	Submission of First Phase Report	
4.	Objective 1	
5.	Objective 2	
6.	Objective 3	
7.	Objective 4	
8.	Testing & Result Analysis	
9.	Documentation of Mini Project Report	

References

- **Maier, A. Sharp, Y . Vagapov** , “Comparative analysis and practical implementation of the ESP32 microcontroller module for the internet of things” 2017 Internet Technologies and Applications, ITA 2017 - Proceedings of the 7th International Conference IEEE, pp 143-148, November 2017, DOI:10.1109/ITECHA.2017.8101926.
- **I. Allafi, T. Iqbal**, “Design and implementation of a Low cost web server using ESP32 for real-time photovoltaic system monitoring” 2017 IEEE Electrical Power and Energy Conference, EPEC 2017, pp 1-5, February 2018, DOI: 10.1109/EPEC.2017.8286184.
- **B.S. Sarjerao, A. Prakasarao**, “A Low Cost Smart Pollution Measurement System Using REST API and ESP32” 3rd International Conference for Convergence in Technology, I2CT 2018, November 2018, DOI: 10.1109/I2CT.2018.8529500.
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Questions?